

MPC-MAP Assignment 5

Week 6: Path Planning

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Task 1 – A* path planning

The path planning algorithm is implemented in `astar.m`. It operates on the grid stored in `read_only_vars.discrete_map.map`. The search maintains an open list sorted by the sum of the cost-so-far g and the heuristic estimate h . The cost of vertical/horizontal moves is 1 and diagonal moves cost $\sqrt{2}$.

The start and goal cells are obtained by discretizing the robot's estimated pose and the map goal with the grid step $\Delta = \text{read_only_vars.map.discretization_step} = 0.2$ m.

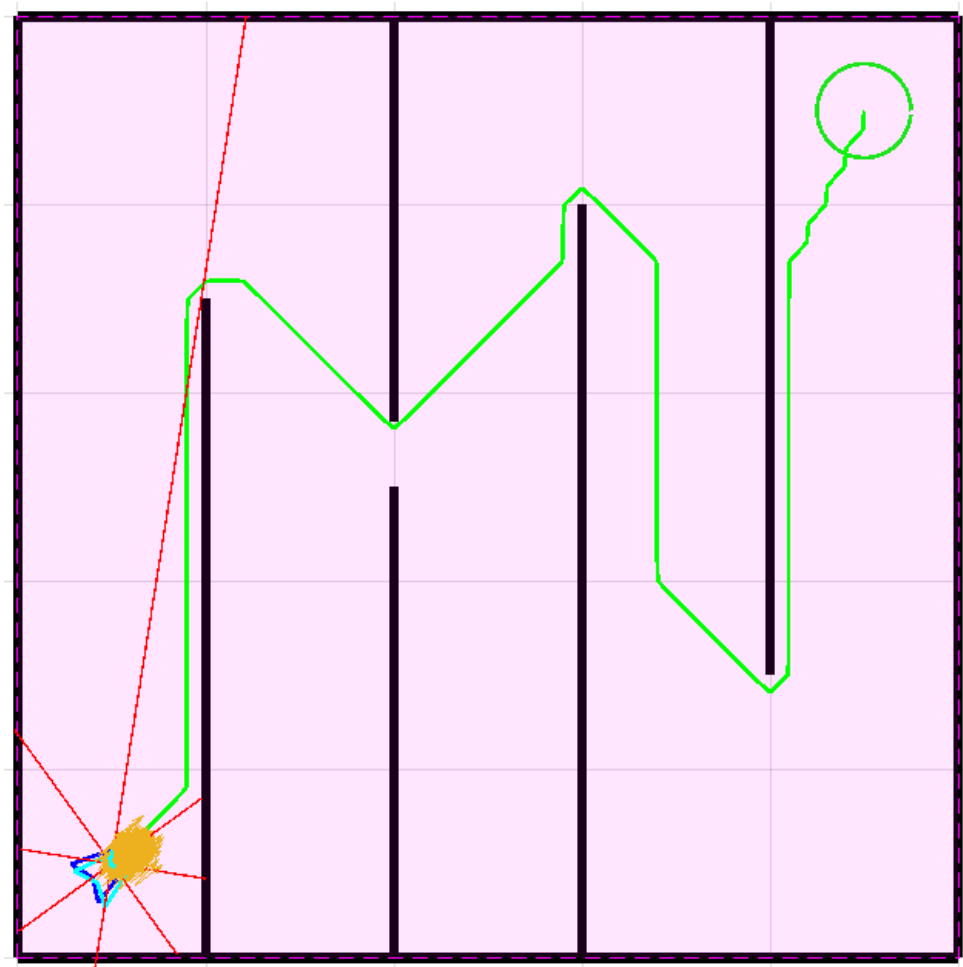


Figure 1: Raw A* path on mixed_1.

Task 2 – Obstacle clearance

Before the search begins the occupancy grid is inflated by one cell around every obstacle. This is done with a nested loop over the grid: for every occupied cell all neighbours are marked as occupied as well. Because the grid step is 0.2 m, the inflation radius is exactly one cell, which guarantees a clearance of at least 0.2 m from any wall. The planner then runs on the inflated grid.

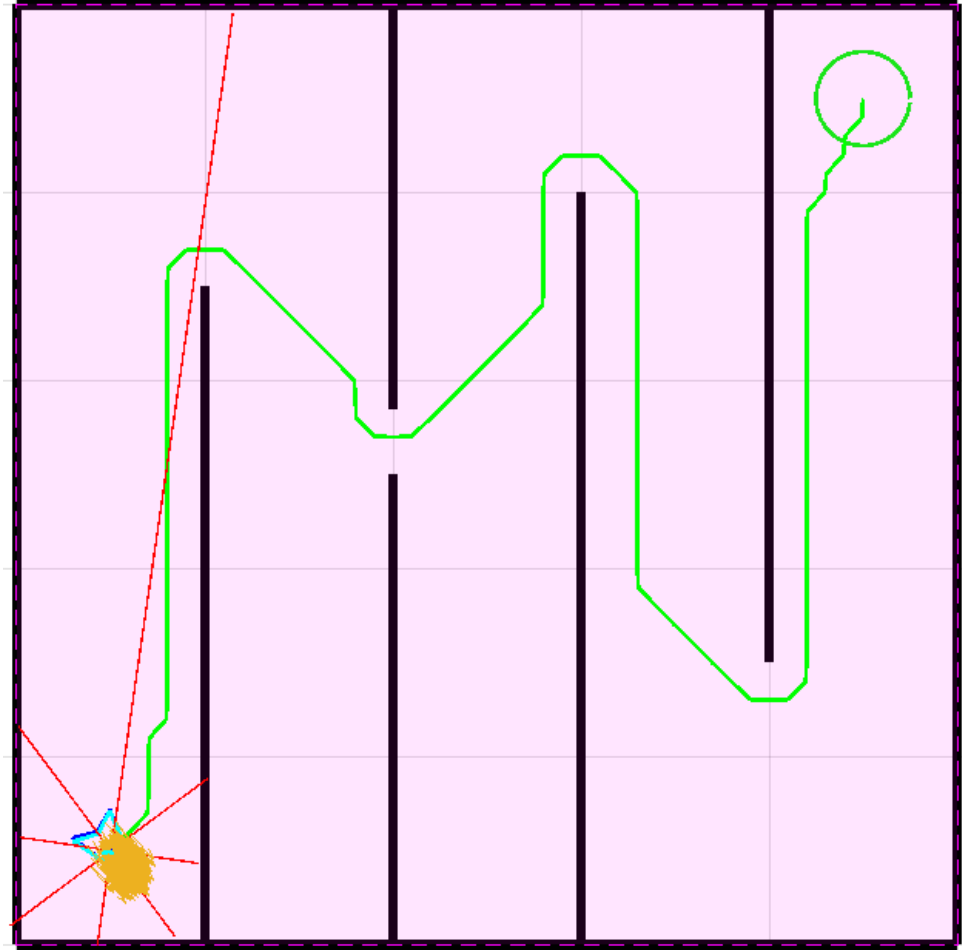


Figure 2: A* path with obstacle inflation on `mixed_1`.

Task 3 – Path smoothing

After the A* search the raw path is passed to `smooth_path.m`, which applies the iterative smoothing algorithm. Each interior point $\mathbf{p}_i$ is updated by two competing terms:

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \alpha (\mathbf{p}_i^{\text{orig}} - \mathbf{p}_i) + \beta (\mathbf{p}_{i-1} + \mathbf{p}_{i+1} - 2\mathbf{p}_i).$$

The first term, weighted by $\alpha = 0.9$ (`weight_data`), pulls the point back toward the original A* solution. The second term, weighted by $\beta = 0.05$ (`weight_smooth`), pulls toward the average of the neighbours and reduces curvature. The iterations stop when either 50 passes are completed or the total change drops below 10^{-4} .

A larger β produces a more rounded trajectory, but the path may cut corners and approach obstacles more closely than the original. A larger α produces a jagged, less natural trajectory.

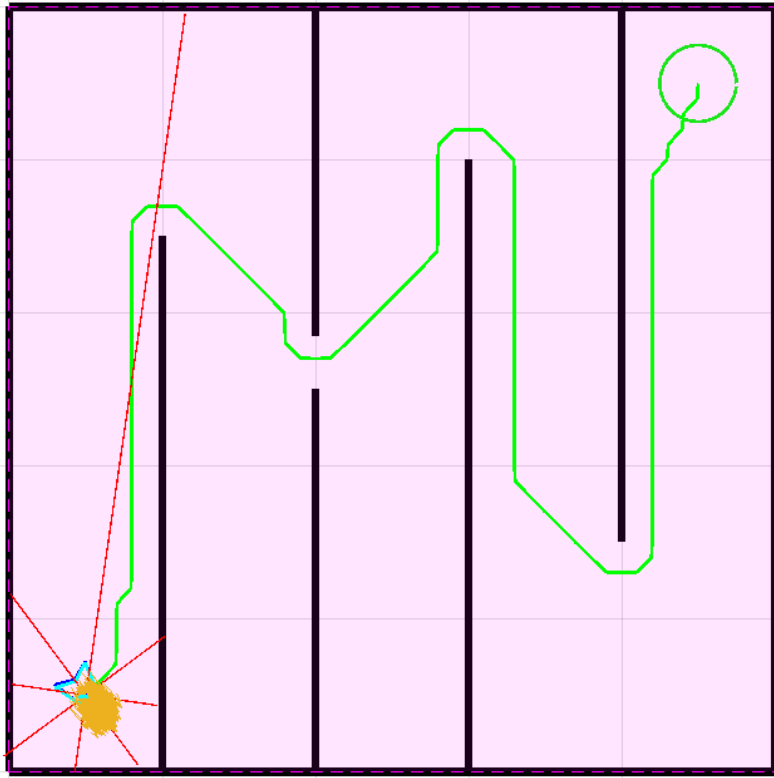


Figure 3: Smoothed path with default parameters $\alpha = 0.9, \beta = 0.05$.

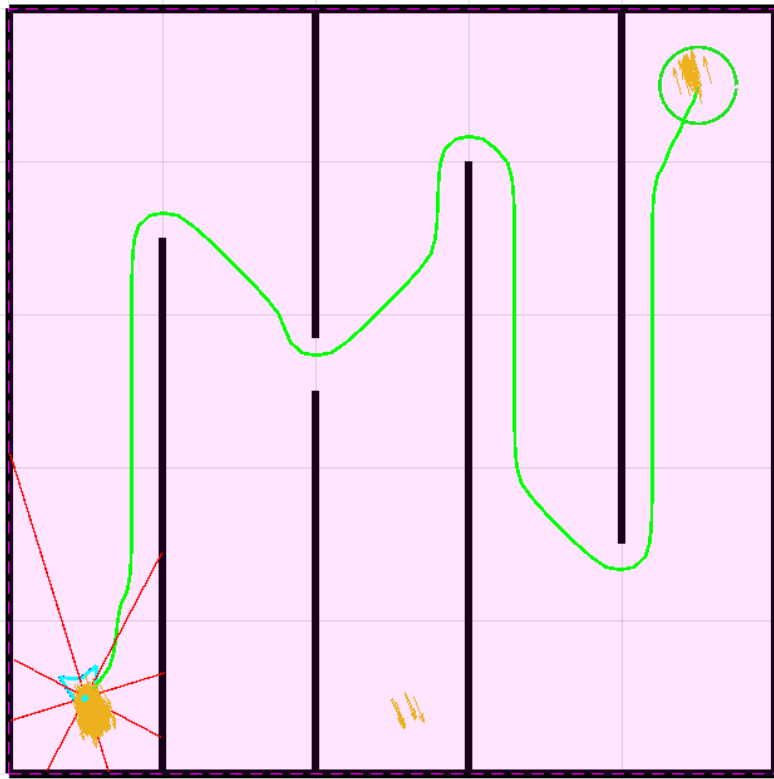


Figure 4: Smoothed path with stronger smoothing $\alpha = 0.5, \beta = 0.2$.